UNIVERSITY OF GHANA

COURSE INFORMATION	
College	College of Basic and Applied Sciences
School / Institute	School of Physical and Mathematical Sciences
Department	Statistics and Actuarial Science
Programme	BSc Actuarial Science
Course Code	ACTU 302
Course Title	Introduction to Actuarial Computing
Credits	Three (3)
Pre-requisites	STAT 111; STAT 112, ACTU 204, STAT223 and STAT 240
Delivery Mode	Face-to-face lectures (2 hrs/week), practical computing sessions (1 hr/week), online resources and demonstrations through Sakai
Academic Year / Semester	2025/2026 | Second Semester

COURSE FACILITATOR				
Facilitator	**Office**	**E-mail**	**Teaching Assistant**	**Lecture Periods**
Prof. Richard Minkah	Room 211 Office Hours: Wednesday 11:30am – 5:30pm	rminkah@ug.edu.gh	Gershon Tetteh Amanor Maame Nyarko Owusu Sekyere	Thursday 9.30am-11.20am Friday: Tutorial/Online class

WELCOME MESSAGE

Dear Students,

Welcome to ACTU 302: Introduction to Actuarial Computing.

This course provides an important transition from understanding the mathematical foundations of actuarial science to implementing actuarial solutions using modern computational tools. As future actuaries, you are expected not only to understand actuarial theory but also to develop the practical skills necessary to solve real-world insurance, pension, investment, and risk management problems.

Throughout the semester, you will learn how to use R and other actuarial software tools to perform simulations, fit actuarial models, analyse financial and insurance data, construct life tables, and implement actuarial techniques used in professional practice. The course combines theoretical concepts with practical applications, preparing you for advanced actuarial work and professional examinations.

Our class operates on three ground rules:

(1) punctuality and active participation during lectures and practical sessions;

(2) submission of assignments and project work by stated deadlines; and

(3) respectful engagement and professional conduct throughout the semester.

Each week combines lectures, practical computing exercises, and online learning resources through Sakai. We look forward to an engaging and productive semester with you.

COURSE DESCRIPTION

ACTU 302 provides a foundation in actuarial computing and introduces students to the computational techniques required in modern actuarial practice. The course serves as a bridge between theoretical actuarial concepts and their practical implementation using statistical and actuarial software.

The course focuses on two major areas of actuarial practice: investment management and life insurance. Students are introduced to programming in R and the use of actuarial packages for data management, visualisation, simulation, estimation, financial modelling, portfolio analysis, survival modelling, life table construction, insurance pricing, and reserving.

Emphasis is placed on reproducible computational methods, interpretation of results, actuarial problem-solving, and professional report writing. Recommended software includes R and Excel.

EXPECTED LEARNING OUTCOMES & INDICATORS

By the end of this course, students should be able to:

- **LO 1 (Knowledge):** Explain the role of computational methods and simulation in actuarial practice.
- **LO 2 (Application):** Write efficient, transparent, and reproducible programs in R for actuarial analysis.
- **LO 3 (Application):** Use actuarial software packages to import, manage, analyse, and visualise actuarial data.
- **LO 4 (Analysis):** Apply simulation methods and Monte Carlo techniques to actuarial problems.
- **LO 5 (Application):** Implement computational solutions for financial mathematics and investment management problems.
- **LO 6 (Analysis):** Construct life tables, fit survival models, and price basic life insurance products using actuarial software.
- **LO 7 (Evaluation):** Interpret computational outputs and communicate findings through professional actuarial reports.

Indicators: Learning outcomes are demonstrated through successful completion of assignments, simulation studies, actuarial projects, practical examinations, coding exercises, and professional written reports.

COURSE ASSESSMENT PLAN			
Component	**Description**	**Weight**	**LO Alignment**
Assignments (x3)	Individual and group practical exercises involving programming, simulations, and actuarial applications	**30%**	LO 1-6
Final Project/IA	Comprehensive actuarial computing project with written report and presentation	**20%**	LO 4-7
Practical Examination	Laboratory-based practical examination involving R programming and actuarial applications	**50%**	LO 1-7

Grading Scale: As per the University of Ghana Undergraduate Handbook.

ACADEMIC INTEGRITY

Academic integrity is central to the UG academic community. All students are required to uphold the principles of honesty, fairness, responsibility, respect, and accountability in all academic work. Plagiarism in any form is a serious offence subject to sanctions stipulated in the University of Ghana Undergraduate Handbook.

Students must familiarise themselves with:

- the UG Plagiarism Policy; and
- the UG Guideline on the Use of Artificial Intelligence.

Unauthorised use of AI-generated content, code, or reports in assessments constitutes academic dishonesty. Any permitted use of AI tools must be appropriately disclosed and acknowledged.

COURSE AND LECTURERS EVALUATION (CLE)

In accordance with the University of Ghana Academic Calendar, course and lecturer evaluations shall be administered during the two-week period immediately preceding the final examinations.

Students are expected to participate actively by providing honest and constructive feedback. The results of these evaluations are used as an important mechanism for quality assurance and the continuous improvement of teaching, learning, and curriculum delivery.

BRIEF CONTENT DESCRIPTION — WEEKLY SCHEDULE		
Week	**Topic**	**Assessment**
1	Introduction and Overview of Actuarial Computing; Introduction to R and Actuarial Packages	
2	Fundamentals of R Programming and Actuarial Packages	

3	Data Visualisation: Univariate and Multivariate Analysis	**Assignment 1**
4	Methods of Generating Random Variables and Monte Carlo Methods	
5	Methods of Estimation: Method of Moments, Maximum Likelihood Estimation and Applications	
6	Simulation Studies in Actuarial Science	**Assignment 2**
7	Financial Mathematics: Cash Flows, Valuation, Pricing, Duration Calculations, Annuities, Asset-Liability Management, Investment Income and Capital Budgeting	
8	Financial Mathematics: Asset Pricing Theory, Portfolio Theory, Mean-Variance Optimisation, Markowitz Two-Asset Portfolio and Introduction to Option Pricing	
9	Life Contingencies: Survival Distributions	**Assignment 3**
10	Life Tables and Pricing of Life Insurance Products	**Project Submission**
11	Reserving Life Insurance Contracts and Group Insurance Applications	
12	Revision and Integrated Applications	
13	Practical Examination in R	**Final Practical Test**

All groups follow the same weekly content schedule. The facilitator delivers all topics across the full 12-week teaching period.

TEACHING AND LEARNING METHODS

The course employs a blended, practice-oriented learning approach:

- **Lectures:** Structured face-to-face lectures introducing actuarial concepts and computational methods.
- **Practical Computing Sessions:** Guided coding demonstrations using R and actuarial software packages.
- **Online Resources:** Supplementary videos, datasets, reading materials, and discussion forums hosted on Sakai.
- **Project-Based Learning:** Students undertake actuarial computing projects involving real-world datasets and applications.
- **Case-Based Examples:** Applications from insurance, pensions, investments, and risk management are used throughout the semester.

LEARNER SUPPORT PLAN

- Weekly office hours are held by the course lecturer.
- Teaching assistant support is available during practical sessions.

- A dedicated Sakai discussion forum is monitored regularly.
- Additional revision sessions may be organised prior to assessments.
- Supplementary computing clinics may be arranged for students requiring additional support.

RESOURCES / TEACHING AND LEARNING MATERIALS

- Lecture notes and presentation slides.
- Practical coding demonstrations and scripts.
- Sample actuarial datasets.
- Weekly practical exercises.
- Project guidelines and report templates.
- Past practical examination papers.
- Youtube videos of class sessions and other recordings

SUPPORTING TECHNOLOGIES

Synchronous:

- Microsoft Teams
- YouTube Live

Asynchronous:

- University of Ghana Learning Management System (Sakai): course materials, assignment submission, grades.
- Discussion Forums: peer and facilitator Q&A.

Other Software:

- R and RStudio
- Microsoft Excel
- Actuarial R Packages (actuar, lifecontingencies, survival, tidyverse)

CORE TEXTS

- Charpentier, A. (2014). *Computational Actuarial Science with* R. Chapman & Hall/CRC.
- Kabacoff, R. I. (2010). *R in Action*. Manning Publications.
- Chambers, J. M. (2008). *Software for Data Analysis: Programming with* R. Springer.
- Bowers, N. L., Gerber, H. U., Hickman, J. C., Jones, D. A., & Nesbitt, C. J. (1997). *Actuarial Mathematics* (2nd ed.). Society of Actuaries.

ADDITIONAL READINGS

- Beckley, J. A., Scahill, P. L., Varitek, M. C., & White, T. A. (2012). *Understanding Actuarial Practice*. Society of Actuaries.

- Dickson, D. C., Hardy, M. R., & Waters, H. R. (2020). *Actuarial mathematics for life contingent risks*. Cambridge University Press.
- Klugman, S. A., Panjer, H. H., & Willmot, G. E. (2012). *Loss models: from data to decisions* (Vol. 715). John Wiley & Sons.

Disclaimer This course outline may be subject to changes as and when necessary. Students will be notified of any amendments via SAKAI. Revised: May 2026 | Department of Statistics & Actuarial Science, University of Ghana, Legon